

Math Bootcamp

Day 3 — Remaining real analysis and linear algebra

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

3.1 Fixed points — equilibrium as a solution

Fixed point. A point x^* is a **fixed point** of a map T if $T(x^*) = x^*$.

Economics reading. Many equilibria are fixed-point problems:

- **Market clearing:** a price p^* with excess demand $E(p^*) = 0$.
- **Game theory:** a strategy profile that is a best response to itself (Nash).

Two useful existence tools (today).

- **Contraction Mapping Theorem:** existence and uniqueness of a fixed point; iteration converges.
- **Brouwer:** continuous map on a closed interval / convex set.

3.1 Contraction Mapping Theorem

Contraction. A map T **shrinks distances** if there is $k \in (0, 1)$ such that $|T(x) - T(y)| \leq k|x - y|$ for all x, y in the domain.

Contraction Mapping Theorem. If $T: [a, b] \rightarrow [a, b]$ is a contraction, then

- **Existence:** there is a fixed point x^* with $T(x^*) = x^*$.
- **Uniqueness:** that fixed point is **unique**.
- **Iteration:** for any starting value $x_0 \in [a, b]$, the sequence $x_n = T(x_{n-1})$ converges to x^* .

3.1 Why iteration converges

Step 1 — Start anywhere. Pick any x_0 . Define $x_n = T(x_{n-1})$ for $n = 1, 2, \dots$

Step 2 — Jumps shrink exponentially. Each new step is at most k times the previous jump:

$$|x_{n+1} - x_n| = |T(x_n) - T(x_{n-1})| \leq k|x_n - x_{n-1}| \leq k^n|x_1 - x_0|.$$

Since $k < 1$, the moves get **exponentially smaller** — the path barely changes after a while.

Step 3 — The sequence converges. Step 2 implies $x_n \rightarrow x^*$ for some limit x^* (check it is a **Cauchy sequence** for a rigorous proof). So iteration from **any** starting point drives x_n toward the same x^* .

Step 4 — The limit is a fixed point. T is continuous, so taking limits in $x_n = T(x_{n-1})$ gives $x^* = \lim x_n = \lim T(x_{n-1}) = T(\lim x_{n-1}) = T(x^*)$.

3.1 Walrasian price adjustment

Market clearing. A price p^* **clears the market** when excess demand is zero: $E(p^*) = 0$. Here $E(p)$ can take **any form** (from supply and demand behind the scenes).

Price adjustment (nonlinear in E). Update the price each period:

$$p_{t+1} = T(p_t) = p_t + \lambda(1 - e^{-\alpha E(p_t)}), \quad \lambda = \frac{1}{5}, \alpha = 0.1.$$

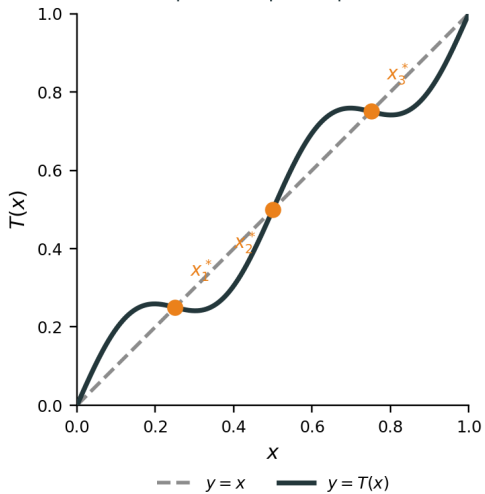
When $|E|$ is small, $1 - e^{-\alpha E} \approx \alpha E$ — move a fraction of excess demand toward balance. Clearing satisfies $E(p^*) = 0 \Leftrightarrow T(p^*) = p^*$. If T is a **contraction** on a price range, then $p_t \rightarrow p^*$ from any starting p_0 .

3.1 Brouwer — one dimension

Brouwer (one dimension). If $T: [a, b] \rightarrow [a, b]$ is **continuous**, then T has **at least one** fixed point x^* with $T(x^*) = x^*$.

Picture. Plot $y = T(x)$ against $y = x$. Any crossing is a fixed point. Continuity forces the graph to meet the diagonal at least once on $[a, b]$.

Multiple fixed points possible



3.1 Brouwer — general statement

Brouwer (several variables). Let $K \subset \mathbb{R}^n$ be **nonempty**, **compact**, and **convex**. If $T: K \rightarrow K$ is **continuous**, then T has at least one fixed point $x^* \in K$ with $T(x^*) = x^*$.

Camp vocabulary.

- **Compact** in $\mathbb{R}^n$: closed and bounded (Heine–Borel).
- **Convex**: the segment between any two points in K stays in K .

One dimension. $K = [a, b]$ is compact and convex, so the 1D case is a special case.

3.1 Brouwer — Cournot duopoly

Two firms. Firm i chooses $q_i \in [0, \bar{q}]$. **Nonlinear** inverse demand $p = a - (q_1 + q_2)^2$ (e.g. $a = 8$, $\bar{q} = 2$).

Profit and FOC. $\pi_i = (a - (q_1 + q_2)^2)q_i$. Firm 1's best response solves

$$a - 3q_1^2 - 4q_1q_2 - q_2^2 = 0 \quad (\text{similarly for firm 2 with } q_1, q_2 \text{ swapped}).$$

The map $q_j \mapsto BR_i(q_j)$ is **nonlinear** — no formula like “ $(a - c - q_j)/2$ ”.

Fixed point = Nash. $T(q_1, q_2) = (BR_1(q_2), BR_2(q_1))$ is continuous on $K = [0, \bar{q}]^2$. **Brouwer** guarantees an equilibrium, but you generally solve $T(q_1^*, q_2^*) = (q_1^*, q_2^*)$ by **algebra or numerics** (here one symmetric solution is $q_1^* = q_2^* = \sqrt{a/8}$).

3.2 Separating Hyperplane Theorem

Setup. Two nonempty **convex** sets $A, B \subset \mathbb{R}^n$ with $A \cap B = \emptyset$.

Theorem. If at least one of A, B is **compact**, then there exist $\mathbf{p} \neq \mathbf{0}$ and $\beta \in \mathbb{R}$ such that $\mathbf{p} \cdot \mathbf{a} \leq \beta \leq \mathbf{p} \cdot \mathbf{b}$ for all $\mathbf{a} \in A, \mathbf{b} \in B$. Geometrically: the hyperplane $\mathbf{p} \cdot \mathbf{x} = \beta$ separates A from B . (Both need not be compact — if only B is compact, swap labels.)

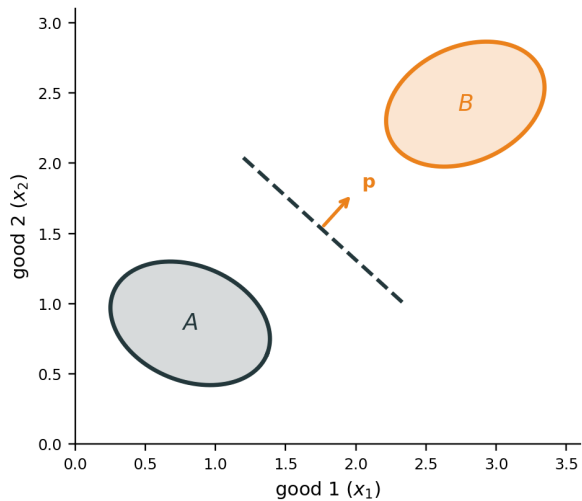
In two goods ($\mathbf{x} = (x_1, x_2)$). The line $p_1x_1 + p_2x_2 = \beta$ puts all of A on one side ($\mathbf{p} \cdot \mathbf{a} \leq \beta$) and all of B on the other ($\mathbf{p} \cdot \mathbf{b} \geq \beta$); $\mathbf{p}$ is the normal, β fixes the line's position.

Remark (single point). $B = \{b_0\}$ is compact. If $b_0 \notin A$ and A is closed and convex, a separating hyperplane exists even when A is not compact.

Remark (strict separation). If the sets are a **positive distance** apart ($d = \inf \|\mathbf{a} - \mathbf{b}\| > 0$; e.g. both compact and disjoint), the same $\mathbf{p}$ gives **strict** separation $\mathbf{p} \cdot \mathbf{a} < \beta < \mathbf{p} \cdot \mathbf{b}$ — **appendix (for interest only)**.

Proof idea when A is compact: **appendix (for interest only)**.

Disjoint convex sets separated by a line



3.2 Example — producer theory (numerical illustration)

Netput plan $y = (y_1, y_2)$: input $y_1 \leq 0$, output $y_2 \geq 0$ (second quadrant).

Technology. With $L = -y_1$ units of labor, max output is $f(L) = 2\sqrt{L}$, so

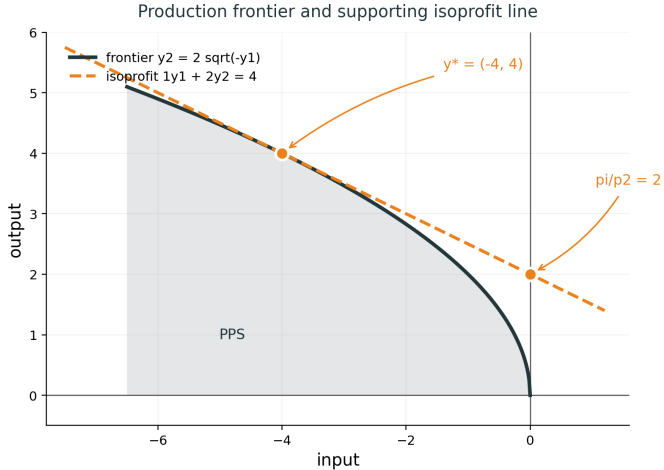
$$Y = \{(y_1, y_2) : y_1 \leq 0, 0 \leq y_2 \leq 2\sqrt{-y_1}\} \quad (\text{free disposal on output}).$$

Prices. Wage $p_1 = 1$, output price $p_2 = 2$. Profit $\pi = p_1 y_1 + p_2 y_2$ (since $y_1 < 0$, the term $p_1 y_1$ is labor cost).

Efficient plan. Use $L = 4$: $y^* = (-4, 4)$. Then $\pi = 1 \cdot (-4) + 2 \cdot 4 = 4$.

Isoprofit line. $y_1 + 2y_2 = 4$, or $y_2 = 2 - \frac{1}{2}y_1$. At $y_1 = 0$: y_2 -axis intercept $y_2 = \pi/p_2 = 2$. Slope $-p_1/p_2 = -\frac{1}{2}$ matches the frontier at y^* (supporting prices).

3.2 Example — producer theory (graph)



3.2 Example — producer theory

Setup. A firm's **production possibility set** $Y \subset \mathbb{R}^n$ is **convex** (and closed in full treatments).

Statement. If Y is convex, the theorem guarantees that every **technically efficient** production plan $y \in Y$ is also a **profit-maximizing** choice for some nonzero price vector $\mathbf{p}$:

$$\mathbf{p} \cdot y \geq \mathbf{p} \cdot y' \quad \text{for all } y' \in Y.$$

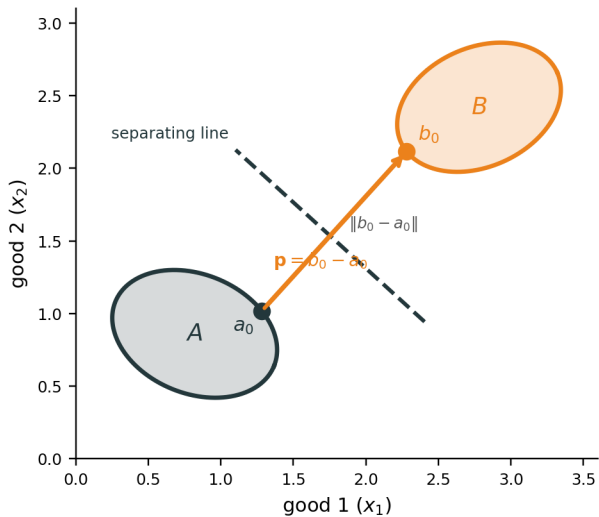
Separation picture. Let B be outputs **strictly better than** y (more of some good, no less of any). If y is efficient, $Y \cap B = \emptyset$. A separating hyperplane gives supporting prices $\mathbf{p}$ at y .

Proof: [appendix](#) (for interest only).

Appendix

Not required for Math Camp. Proofs and extra examples linked from the main slides.

Proof idea: closest pair, then a line $\perp \mathbf{p}$



A.1 Proof. Separation when A is compact

for interest only

Claim. Disjoint nonempty convex $A, B \subset \mathbb{R}^n$ with A compact and B closed. Then $\exists \mathbf{p} \neq \mathbf{0}, \beta \in \mathbb{R}$ with $\mathbf{p} \cdot \mathbf{a} \leq \beta \leq \mathbf{p} \cdot \mathbf{b}$ for all $\mathbf{a} \in A, \mathbf{b} \in B$.

Step 1 — closest pair. Define $d = \inf\{\|\mathbf{a} - \mathbf{b}\| : \mathbf{a} \in A, \mathbf{b} \in B\} > 0$. Pick sequences $(\mathbf{a}_n) \subset A, (\mathbf{b}_n) \subset B$ with $\|\mathbf{a}_n - \mathbf{b}_n\| \rightarrow d$. Since A is compact, a subsequence $\mathbf{a}_{n_k} \rightarrow \mathbf{a}_0 \in A$. The $(\mathbf{b}_{n_k})$ are bounded, so (using B closed) a further subsequence $\mathbf{b}_{n_{k_j}} \rightarrow \mathbf{b}_0 \in B$. Hence the infimum is **attained** at some pair $(\mathbf{a}_0, \mathbf{b}_0) \in A \times B$.

Not an interior point. $\mathbf{a}_0$ and $\mathbf{b}_0$ need only lie in A and B ; they are usually on the **boundaries**, not in $\text{int}(A)$ or $\text{int}(B)$.

Step 2 — normal vector. Set $\mathbf{p} = \mathbf{b}_0 - \mathbf{a}_0 \neq \mathbf{0}$ (points from $\mathbf{a}_0$ toward $\mathbf{b}_0$).

A.1 Step 1 — why the subsequence argument works

for interest only

Why $a_{n_k} \rightarrow a_0 \in A$. In $\mathbb{R}^n$, **compact** = closed + bounded (Heine–Borel). Every sequence in a compact set has a **subsequence** converging to a limit **in that set**. So $(a_n) \subset A$ compact $\Rightarrow \exists a_{n_k} \rightarrow a_0 \in A$.

Why (b_{n_k}) is bounded. Since $\|a_{n_k} - b_{n_k}\| \rightarrow d$ and (a_{n_k}) converges, for large k ,

$$\|b_{n_k}\| \leq \|b_{n_k} - a_{n_k}\| + \|a_{n_k}\| \leq d + 1 + \|a_0\| + 1.$$

(Triangle inequality: $\|b\| \leq \|b - a\| + \|a\|$.)

Why $b_{n_{k_j}} \rightarrow b_0 \in B$. A bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (Bolzano–Weierstrass). Since B is **closed**, the limit b_0 lies in B .

A.1 Step 3 — why the derivative works

for interest only

Setup. Fix $a \in A$. For $t \in [0, 1]$, convexity gives $a_t = a_0 + t(a - a_0) \in A$ and $g(t) = \|a_t - b_0\|^2$. Since (a_0, b_0) minimizes distance, $t = 0$ minimizes g on $[0, 1]$.

Why use the derivative? A minimum at the left endpoint $t = 0$ means the slope cannot point downward: $g'(0) \geq 0$.

Differentiate. $g(t) = \|a_0 + t(a - a_0) - b_0\|^2$, so

$$g'(t) = 2\langle a_0 + t(a - a_0) - b_0, a - a_0 \rangle, \quad g'(0) = 2\langle a_0 - b_0, a - a_0 \rangle \geq 0.$$

Hence $\langle b_0 - a_0, a - a_0 \rangle \leq 0$. With $\mathbf{p} = b_0 - a_0$: $\mathbf{p} \cdot (a - a_0) \leq 0$, so $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$.

A.1 Proof. Separation when A is compact (continued)

Step 4 — B stays on the other side. Fix $b \in B$. For $t \in [0, 1]$, set $b_t = b_0 + t(b - b_0) \in B$ and $h(t) = \|a_0 - b_t\|^2$. Since $t = 0$ minimizes h on $[0, 1]$, the same derivative argument gives $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$ for all $b \in B$.

Step 5 — finish. Since $A \cap B = \emptyset$, the closest points satisfy $a_0 \neq b_0$. With $\mathbf{p} = b_0 - a_0$,

$$\mathbf{p} \cdot b_0 - \mathbf{p} \cdot a_0 = \mathbf{p} \cdot (b_0 - a_0) = \|b_0 - a_0\|^2 = d^2 > 0,$$

so $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$. Any β with $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$ works (e.g. $\beta = \mathbf{p} \cdot a_0$ gives **weak** separation: $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A, b \in B$).

A.1 Strict separation

for interest only

Weak vs. strict. From Steps 3–5: $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$ and $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$. So any β with $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$ separates weakly (e.g. $\beta = \mathbf{p} \cdot a_0$).

When is separation strict? If $d = \|b_0 - a_0\| > 0$ (sets bounded apart; e.g. both compact and disjoint), then $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$ with a **positive gap**. Pick β **strictly between** them, e.g. $\beta = \frac{1}{2}(\mathbf{p} \cdot a_0 + \mathbf{p} \cdot b_0)$. Then

$$\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0 < \beta < \mathbf{p} \cdot b_0 \leq \mathbf{p} \cdot b \quad \Rightarrow \quad \mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b.$$

If $d = 0$ (sets touch in the limit, e.g. $A = [0, 1]$, $B = (1, 2]$), only weak separation is possible.

A.1 Proof. Remarks

for interest only

If only B is compact. Swap the roles of A and B , apply the proof, then relabel the normal vector.

Single point $B = \{b_0\}$. Let $\hat{a}$ be a point in A closest to b_0 (exists since A is closed and convex). The same calculation with $(a_0, b_0) = (\hat{a}, b_0)$ separates A from b_0 even when A is not compact.

A.2 Proof. Efficient production $\Rightarrow$ profit maximization

for interest only

Goal. $y \in Y$ technically efficient $\Rightarrow \exists \mathbf{p} \neq \mathbf{0}$ with $\mathbf{p} \cdot y \geq \mathbf{p} \cdot y'$ for all $y' \in Y$.

Assumptions (camp level). $Y \subset \mathbb{R}^n$ is **closed and convex**; Y is **bounded** (so compact); **free disposal** (if $y \in Y$ and $\tilde{y} \leq y$, then $\tilde{y} \in Y$).

Step 1 — better set. Define

$$P_y = \{y' \in \mathbb{R}^n : y' \geq y, y' \neq y\}$$

(weakly more output, strictly more in at least one good). P_y is **convex**.

Step 2 — disjointness. If $y' \in P_y \cap Y$, then y' weakly dominates y in Y , contradicting efficiency. So $P_y \cap Y = \emptyset$.

A.2 Proof. Efficient production (continued)

for interest only

Step 3 — separate. Y is compact and P_y is closed, so apply the separating hyperplane theorem (swap labels if needed so a compact set is on one side). There exist $\mathbf{p} \neq \mathbf{0}$ and β with

$$\mathbf{p} \cdot \mathbf{y}'' \leq \beta \leq \mathbf{p} \cdot \mathbf{y}' \quad \forall \mathbf{y}'' \in Y, \forall \mathbf{y}' \in P_y.$$

Hence $\mathbf{p} \cdot \mathbf{y}'' \leq \mathbf{p} \cdot \mathbf{y}'$ for all $\mathbf{y}'' \in Y, \mathbf{y}' \in P_y$.

Step 4 — profit maximization at y . For each good i , pick $\mathbf{y}^{(i)} = y + \varepsilon \mathbf{e}_i \in P_y$ for small $\varepsilon > 0$ (feasible with free disposal). Then $\mathbf{p} \cdot \mathbf{y}'' \leq \mathbf{p} \cdot \mathbf{y}^{(i)} = \mathbf{p} \cdot y + \varepsilon p_i$. Let $\varepsilon \downarrow 0$ to get $\mathbf{p} \cdot \mathbf{y}'' \leq \mathbf{p} \cdot y$ for all $\mathbf{y}'' \in Y$.

Remark. Separation gives only $\mathbf{p} \neq \mathbf{0}$. For $\mathbf{p} \geq 0$, need free disposal on all goods: if $p_i < 0$, then $y - \varepsilon \mathbf{e}_i \in Y$ (since $\leq y$) raises profit, contradicting maximization at y .